

ANSWER KEY

Mid-term test ML

Course: CSC354 - Machine Learning

CLO-2 Apply supervised learning techniques to solve classification problems.

Q1. Explain how you would use k-nearest neighbors (KNN) to classify new data points in a dataset with two features.

Answer:

To classify new data points using KNN, first determine the value of k . Then find the k nearest training samples based on their feature distances from the new point. Assign the class label by majority vote among these neighbors.

Marks: 5

Q2. Explain how you would use logistic regression to classify emails as spam or not spam using supervised learning techniques.

Answer:

Logistic regression can be used for binary classification problems like email spam detection. You would first collect a labeled dataset of emails, where each email is tagged with whether it is spam (1) or not spam (0). Then, you preprocess the data by converting text into numerical features such as word counts or using TF-IDF vectors. After feature extraction, split the dataset into training and testing sets. Train the logistic regression model on the training set to learn the relationship between email content and its label. Finally, evaluate the model's performance on the test set using metrics like accuracy, precision, recall, and F1 score.

Marks: 5

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